

OpenStax Calculus Volume 3

Section 4.2 Exercises: Limits and Continuity

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For the following exercises, find the limit of the function.

1. $\lim_{(x,y) \rightarrow (1,2)} x$
2. $\lim_{(x,y) \rightarrow (1,2)} \frac{5x^2y}{x^2+y^2}$
3. Show that the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{5x^2y}{x^2+y^2}$ exists and is the same along the paths: y -axis and x -axis, and along $y = x$.

For the following exercises, evaluate the limits at the indicated values of x and y . If the limit does not exist, state this and explain why the limit does not exist.

4. $\lim_{(x,y) \rightarrow (0,0)} \frac{4x^2+10y^2+4}{4x^2-10y^2+6}$
5. $\lim_{(x,y) \rightarrow (11,13)} \sqrt{\frac{1}{xy}}$
6. $\lim_{(x,y) \rightarrow (0,1)} \frac{y^2 \sin x}{x}$
7. $\lim_{(x,y) \rightarrow (0,0)} \sin\left(\frac{x^8+y^7}{x-y+10}\right)$
8. $\lim_{(x,y) \rightarrow (\pi/4,1)} \frac{y \tan x}{y+1}$
9. $\lim_{(x,y) \rightarrow (0,\pi/4)} \frac{\sec x+2}{3x-\tan y}$
10. $\lim_{(x,y) \rightarrow (2,5)} \left(\frac{1}{x} - \frac{5}{y}\right)$
11. $\lim_{(x,y) \rightarrow (4,4)} x \ln y$
12. $\lim_{(x,y) \rightarrow (4,4)} e^{-x^2-y^2}$
13. $\lim_{(x,y) \rightarrow (0,0)} \sqrt{9-x^2-y^2}$
14. $\lim_{(x,y) \rightarrow (1,2)} (x^2y^3 - x^3y^2 + 3x + 2y)$
15. $\lim_{(x,y) \rightarrow (\pi,\pi)} x \sin\left(\frac{x+y}{4}\right)$
16. $\lim_{(x,y) \rightarrow (0,0)} \frac{xy+1}{x^2+y^2+1}$
17. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2+y^2}{\sqrt{x^2+y^2+1}-1}$

18. $\lim_{(x,y) \rightarrow (0,0)} \ln(x^2 + y^2)$

For the following exercises, complete the statement.

19. A point (x_0, y_0) in a plane region R is an interior point of R if _____.

20. A point (x_0, y_0) in a plane region R is called a boundary point of R if _____.

For the following exercises, use algebraic techniques to evaluate the limit.

21. $\lim_{(x,y) \rightarrow (2,1)} \frac{x-y-1}{\sqrt{x-y}-1}$

22. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - 4y^4}{x^2 + 2y^2}$

23. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x - y}$

24. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$

For the following exercises, evaluate the limits of the functions of three variables.

25. $\lim_{(x,y,z) \rightarrow (1,2,3)} \frac{xz^2 - y^2z}{xyz - 1}$

26. $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{x^2 - y^2 - z^2}{x^2 + y^2 - z^2}$

For the following exercises, evaluate the limit of the function by determining the value the function approaches along the indicated paths. If the limit does not exist, explain why not.

27. $\lim_{(x,y) \rightarrow (0,0)} \frac{xy + y^3}{x^2 + y^2}$

(a) Along the x -axis ($y = 0$)

(b) Along the y -axis ($x = 0$)

(c) Along the path $y = 2x$

28. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{xy + y^3}{x^2 + y^2}$ using the results of previous problem.

29. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^4 + y^2}$

(a) Along the x -axis ($y = 0$)

(b) Along the y -axis ($x = 0$)

(c) Along the path $y = x^2$

30. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^4 + y^2}$ using the results of previous problem.

Discuss the continuity of the following functions. Find the largest region in the xy -plane in which the following functions are continuous.

31. $f(x, y) = \sin(xy)$

32. $f(x, y) = \ln(x + y)$

33. $f(x, y) = e^{3xy}$

34. $f(x, y) = \frac{1}{xy}$

For the following two exercises, determine the region in which the function is continuous. Explain your answer.

35. $f(x, y) = \frac{x^2 y}{x^2 + y^2}$

36. $f(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}$

37. Determine whether $f(x, y)$ is continuous at $(0, 0)$. $f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$

38. Determine whether $g(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$ is continuous at $(0, 0)$.

39. Create a plot using graphing software to determine where the limit does not exist. Find where in the coordinate plane $f(x, y) = \frac{1}{x^2 - y}$ is continuous.

40. Determine the region of the xy -plane in which the function $g(x, y) = \arctan\left(\frac{xy^2}{x+y}\right)$ is continuous. Use technology to support your conclusion.

41. Determine the region of the xy -plane in which $f(x, y) = \ln(x^2 + y^2 - 1)$ is continuous. Use technology to support your conclusion. (Hint: Choose the range of values for x and y carefully!)

42. At what points in space is $g(x, y, z) = x^2 + y^2 - 2z^2$ continuous?

43. At what points in space is $g(x, y, z) = \frac{1}{x^2 + z^2 - 1}$ continuous?

44. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{1}{x^2 + y^2}$ does not exist at $(0, 0)$ by plotting the graph of the function.

45. [T] Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{-xy^2}{x^2 + y^4}$ by plotting the function using a CAS. Determine analytically the limit along the path $x = y^2$.

46. [T]

(a) Use a CAS to draw a contour map of $z = \sqrt{9 - x^2 - y^2}$.

(b) What is the name of the geometric shape of the level curves?

(c) Give the general equation of the level curves.

(d) What is the maximum value of z ?

(e) What is the domain of the function?

(f) What is the range of the function?

47. True or False: If we evaluate $\lim_{(x,y) \rightarrow (0,0)} f(x)$ along several paths and each time the limit is 1, we can conclude that $\lim_{(x,y) \rightarrow (0,0)} f(x) = 1$.

48. Use polar coordinates to find $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin \sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}}$. You can also find the limit using L'Hôpital's rule.

49. Use polar coordinates to find $\lim_{(x,y) \rightarrow (0,0)} \cos(x^2 + y^2)$.

50. Discuss the continuity of $f(g(x, y))$ where $f(t) = 1/t$ and $g(x, y) = 2x - 5y$.

51. Given $f(x, y) = x^2 - 4y$, find $\lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$.

52. Given $f(x, y) = x^2 - 4y$, find $\lim_{h \rightarrow 0} \frac{f(1+h, y) - f(1, y)}{h}$.